

Architecture Selection Justification

Architecture Selection: "We chose Microservices Architecture for the EV Charging Station System."

Architectural Choice: Microservices Architecture

Two Reasons:

1. Independent Scaling for Peak Loads: The system must maximize resource utilization during peaks (NFR-001) and calculate dynamic power rates based on grid load (FR-03). Microservices allow us to independently scale the Order Manager and Station Controller to handle high traffic without scaling the entire system.
2. Fault Isolation and Modularity: Real-time charging socket monitoring (FR-05) and payment processing (FR-04) are distinct concerns. By separating them into independent services (Order Manager Component and Payment Service Component), an issue with real-time tracking won't bring down the payment gateway.

Security Advantage:

Microservices provide strict boundaries between components. The Payment Service Component can be isolated in a highly secure environment to protect saved credit card information (Epic 3), ensuring that even if the User Interface or Order Manager is compromised, the financial data remains secure.

Performance Benefit:

By independently deploying and load-balancing the Order Manager Component, the system can ensure fast response times, meeting the non-functional requirement to respond to user search queries for charging stations within 2 seconds (NFR-002).